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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2023-2024

Program/s: MBATech/BTech

Year: II Semester: III

Stream/s : Data Science

Subject: Managing Uncertainty

Time: 03 hrs (10:00am to 1:00pm)

Date: 08 /02 / 2024

No. of Pages: 05

Marks: 100

Re-Examination (2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1 CO- 1; SO- 1; BL-2	a.	Explain differences between classification and tabulation.	[5]
CO- 1; SO-1 ; BL-1	b.	Explain dot plots with suitable example.	[5]
CO-2 ; SO-2 ; BL-3	c.	On average, every one out of 10 telephones is found busy. Six telephone numbers are selected at random. Find the probability that four of them will be busy.	[5]
CO- 3; SO- 2; BL-2	d.	For the given observations for random variables X and Y, calculate values for the following transformations. E(X)=15, V(X)=4, E(Y)=30, V(Y)=2 a) E(4Y-8) b) E(X-Y) c) V(X-Y) d) V(4X+8)	[5]

Q2 CO-1; SO-2; BL-5 — a. — [10]

From the following data:

Class	Frequency
20-29	6
30-39	16
40-49	21
50-59	29
60-69	25
70-79	22
80-89	11
90-99	7
100-109	4
110-119	0
120-129	2

Compute Sample Mean using the Step-deviation/Coding method with 0 assigned to the 60-69 class.
Instruction – Do not make classes continuous.

CO-2 ; SO-2 ; BL-5 — b. — [10]

Calculate Spearman's rank correlation coefficient from the following data:

x	10	12	8	15	20	25	40
y	15	10	6	25	16	12	8

Comment and explain the type of relationship that variables x and y exhibit from your answers.

Q3 CO-2; SO-2; BL-3 — a. — [10]

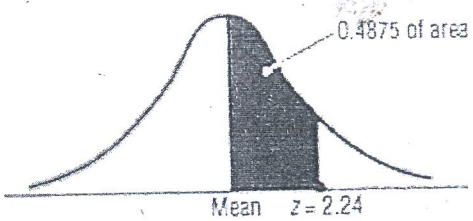
The number of checks cashed each day at five branches of the Bank of Orange County during the past month had the following frequency distribution:

Class	Frequency
0-199	10
200-399	13
400-599	17
600-799	42
800-999	18

Hank Spivey, director of operations for the bank, knows that a standard deviation in check cashing of more than 200 checks per day creates staffing and organizational problems at the branches because of uneven workload. Should Hank worry about staffing next month?

CO-2 ; SO- 6; BL-3	b.	A box contains three coins: two regular coins and one fake two-headed coin (P(H)=1), a) You pick a coin at random and toss it. What is the probability that it lands heads up? b) You pick a coin at random and toss it, and get heads. What is the probability that it is the two-headed coin?	[10]
Q4 CO-2; SO-6; BL-2	a.	Three identical boxes contain red and white balls. The first box contains 3 red and 2 white balls, the second box has 4 red and 5 white balls, and the third box has 2 red and 4 white balls. A box is chosen very randomly and a ball is drawn from it. If the ball that is drawn out is red, what will be the probability that the second box is chosen?	[5]
CO-3; SO-6; BL-3	b.	The total duration of baseball games in the major league in the 2011 season is uniformly distributed between 447 hours and 521 hours inclusive. • Find the mean. • What is the probability that the duration of games for a team for the 2011 season is between 480 and 500 hours?	[5]
CO-3; SO-2; BL-3	c.	Assume that you usually get 2 phone calls per hour. calculate the probability, that a phone call will come within the next hour. Also, calculate the expected value and standard deviation for the given distribution.	[10]
Q5 CO-2; SO-2; BL-2	a.	Find the variance of the following data table **Class** **Frequency** 0-10 3 10-20 6 20-30 4 30-40 2 40-50 1	[10]
CO- 3; SO- 6; BL-5	b.	In Lucknow Railway Station, recently there was a complaint about the accidents on its elevators and the resulting injuries, especially among the elderly people. According to past records, on average, five people met with	[10]

		an accident every week. The station in charge has requested the finance department to allocate extra funds for purchasing safety devices to check the accidents on the elevators. The finance department has replied that funds cannot be allocated unless the chances of more than three accidents on elevators in any week exceed 70 percent. Will funds be allocated?	
Q6 CO-3; SO-2; BL-3	a.	How PMF of a function of a random variable X can be generated using the PMF of X. Let X be a discrete random variable with PX(k)=1/4 for k=−1,0,1,2. Let $Y=X^2$. Find the range and PMF of Y.	[10]
CO-3 ; SO- 1; BL-1	b.	Describe the concept of PDF and CDF for a continuous random variable with a suitable example.	[10]
Q7 CO-3; SO-6; BL-5	a.	The length of human pregnancies from conception to birth approximates a normal distribution with a mean of 266 days and a standard deviation of 16 days. What proportion of all pregnancies will last between 240 and 270 days (roughly between 8 and 9 months)?	[10]
CO- 3; SO- 6; BL-5	b.	Two independent samples of observations were collected. For the first sample of 60 elements, the mean was 86 and the standard deviation 6. The second sample of 75 elements had a mean of 82 and a standard deviation of 9. (a) Compute the estimated standard error of the difference between the two means. (b) Using $\alpha = 0.01$, test whether the two samples can reasonably be considered to have come from populations with the same mean.	[10]



EXAMPLE: TO FIND THE AREA UNDER THE CURVE BETWEEN THE MEAN AND A POINT 2.24 STANDARD DEVIATIONS TO THE RIGHT OF THE MEAN, LOOK UP THE VALUE OPPOSITE 2.2 AND UNDER 0.04 IN THE TABLE; 0.4875 OF THE AREA UNDER THE CURVE LIES BETWEEN THE MEAN AND A z VALUE OF 2.24.

APPENDIX TABLE 1 AREAS UNDER THE STANDARD NORMAL PROBABILITY DISTRIBUTION BETWEEN THE MEAN AND POSITIVE VALUES OF z

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990